

Deblurring Images while Maintaining Emotion

Noah Scott and Dylan Wong

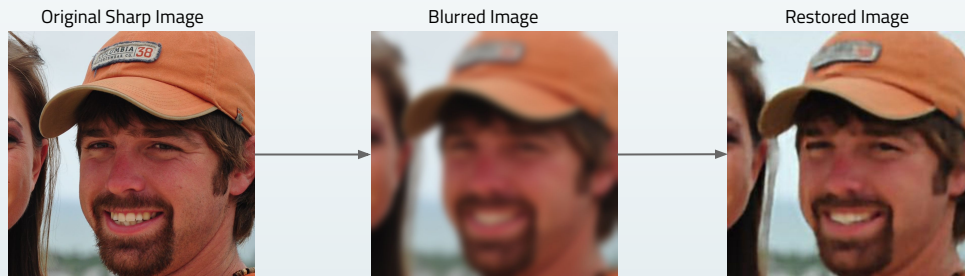
Abstract

This project explores the challenge of restoring blurred facial images while preserving the subject's emotional expression, which is a detail commonly lost in standard deblurring approaches. We use a combined pipeline consisting of a CNN trained for image deblurring and a separate machine learning model trained for emotion classification. Using a dataset of over 70,000 high resolution images, we generated controlled blur distortions and trained an encoder-decoder CNN to reconstruct sharper faces. We then evaluated the restored outputs using a pretrained emotion recognition model to ensure emotional fidelity. Our results show that it is possible to recover both image clarity and emotional expression, demonstrating a promising approach for real world applications such as surveillance, and authentication.

Introduction

Facial deblurring restores blurry or distorted facial images to a recognizable form. This is useful in real world scenarios such as improving security footage or enhancing face based authentication, where clarity is essential for identification. However, many restoration models focus only on recovering sharpness and texture, and often fail to preserve the subtle emotional cues in the original image. Since emotion is a core part of human expression, losing it limits the practical value of these models. In this project, we aim to restore image quality while also maintaining the subject's emotional integrity. We combine a convolutional neural network with a pretrained emotion recognition model to see whether the restored faces retain their original expressive features. This dual model approach allows us to measure both image reconstruction and emotional consistency, leading to more reliable and human aligned facial restoration.

Results



Methodology

We used the Flickr-Faces-HQ (FFHQ) dataset to obtain high-quality facial images for training. After reducing the dataset size by removing unnecessary samples, we applied a Hugging Face emotion-classification model to label each image. We then generated blurred versions of the images by applying Gaussian blur, creating thousands of blurred-sharp pairs for supervised training. These blurred images were used to train our CNN models, including our encoder-decoder network, shallow CNN, shallow residual network, and dilated CNN. Each model learned to map blurred inputs to sharp outputs by extracting features through convolutional layers and minimizing L1 loss between its predictions and the ground-truth images. After training, we again ran the Hugging Face emotion model on the restored images and compared the predicted emotions before and after deblurring.

Conclusion

In conclusion, our experiments showed clear performance differences between the four CNN architectures. The regular encoder-decoder CNN achieved the best results, with the highest PSNR and SSIM scores and the lowest MSE and L1 loss, while the shallow CNN performed the worst due to its limited capacity. The shallow residual and dilated models offered moderate improvements but still fell short of the full encoder-decoder design. Overall, the regular CNN consistently produced the sharpest and most detailed reconstructions, recovering key facial features from heavily blurred inputs. Although its loss stabilized around 0.02, further training or architectural refinement may lower it. Future work could incorporate emotion-aware training to better preserve both image quality and facial expression information.

Model Architecture

We implemented four CNN-based deblurring models in PyTorch: a regular encoder-decoder CNN, a shallow CNN, a shallow residual CNN, and a dilated CNN. The encoder-decoder model consisted of three convolutional layers that progressively extracted features from the blurred image, followed by symmetric convolutional layers that reconstructed the sharpened output. The shallow CNN used only two convolutional layers, while the shallow residual model added a skip connection so the network learned a residual correction added back to the input. The dilated CNN increased its receptive field by using a dilated convolution in the second layer. All models used ReLU activations and were trained with L1Loss to measure pixel-wise error between the deblurred and ground truth images. Each model was trained for 20 epochs on a dataset containing thousands of blurred-sharp image pairs.

Acknowledgements

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